

1. Thesis Overview

1.1 Thesis Identity

Full Title	A Novel Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification
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Supervisor	Dr. Baraa Ismeal Farhan — Assistant Professor
Institution	Wasit University — College of Education for Pure Sciences — Dept. of Computer
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Year	2026 A.D. / 1447 A.H.
Domain	Deep Learning Medical Imaging Brain Tumor Classification Edge AI

1.2 Abstract

Brain tumor classification from MRI images is a critical task in clinical neuro-oncology. This thesis presents a novel ultra-lightweight CNN model designed for low-power clinical and edge environments. The proposed model integrates three intelligent modules: (1) Micro Adaptive Feature Extractor (MAFE) for dynamic spatial and channel feature refinement; (2) Micro Multi-Scale Processor (MMSP) for combining multi-resolution features; and (3) Context-Aware Intelligent Pooling (CAIP) for content-adaptive pooling to improve classification accuracy.

The model is trained and tested on the Mendeley Brain Tumor MRI dataset (12,064 high-resolution images — four classes: glioma, meningioma, pituitary adenoma, and no tumor), achieving 99.34% accuracy with only 659,228 trainable parameters — 80% fewer than comparable U-Net-based methods. Cross-dataset generalization was validated on three external datasets: Kaggle (99.24%), Br35H (98.33%), and Figshare (87.47%). Clinical deployment metrics: 12 ms inference latency, 2.6 MB model size, 4.4 MB RAM usage.

2. Novelty Analysis — Why "Novel" Belongs in the Title

2.1 The Three Research Gaps Addressed

The thesis identifies and addresses three specific gaps that no prior published work has simultaneously resolved:

Gap	Problem in Prior Art	Solution in This Thesis
Accuracy-Efficiency Gap	No architecture achieves >99% accuracy AND sub-3 MB model size for mobile deployment. Best: Khan & Auvée (2024) at 98.67% / 1.1M params.	MAFE + MMSP + CAIP achieve 99.34% accuracy with only 659K parameters and 2.6 MB — both targets surpassed simultaneously.
Attention Design Gap	Prior methods use sequential (CBAM) or channel-only (SE) attention. No parallel dual-pathway attention co-designed within a lightweight budget.	MAFE employs parallel spatial + channel attention in a single step with only ~8K parameters — architecturally distinct from CBAM and SE Networks.
Adaptive Pooling Gap	All lightweight architectures use fixed max/average pooling. No sigmoid-gated content-adaptive pooling within a sub-700K parameter budget.	CAIP introduces sigmoid-gated adaptive blending of max and average pooling, contributing alone +2.18% accuracy improvement (ablation study).

2.2 The Three Novel Architectural Modules

MAFE — Micro Adaptive Feature Extractor

Purpose: Replaces standard convolutional feature extraction with a dual-pathway spatial + channel attention mechanism operating in parallel (unlike CBAM's sequential design).

Novelty: Uses only ~8K parameters to achieve simultaneous focus on tumor region location (spatial) and feature importance (channel), which prior lightweight models could not do within a sub-700K budget.

Distinction from prior art: SE Networks (Hu et al., 2018) perform channel-only recalibration; CBAM (Woo et al., 2018) performs spatial and channel attention sequentially — neither achieves the parallel co-design of MAFE.

MMSP — Micro Multi-Scale Processor

Purpose: Processes features at multiple scales (fine, boundary, global) simultaneously — mimicking how an experienced radiologist examines tumors at different resolutions.

Novelty: Incorporates a depthwise separable convolution (DWConv) pathway and a learnable fusion mechanism, making true multi-scale processing feasible within a sub-700K architecture. Prior multi-scale methods (e.g., FPN) required full encoder-decoder structures.

CAIP — Context-Aware Intelligent Pooling

Purpose: Replaces static global average/max pooling with a sigmoid-gated content-adaptive pooling that dynamically balances boundary details vs. noise suppression based on image content.

Novelty: First integration of learned sigmoid-gated adaptive blending within a sub-700K parameter lightweight architecture for medical imaging. Ablation study confirms +2.18% accuracy contribution from CAIP alone.

2.3 Performance Evidence Supporting Novelty

Model	Accuracy	Parameters	Key Limitation
This Thesis (Modified CNN)	99.34%	659K (2.6 MB)	Image-level (not patient-level) split
Khan et al. (2025)	99.11%	>10M	Transfer learning dependency
Zahoor et al. (2024)	98.83%	~8.2M	Heavy model, not deployable
Nazir et al. (2024)	98.67%	—	Binary classification only
Khan & Auvee (2024)	98.67%	~1.1M	Accuracy below 99%
Tummala et al. (2022)	98.70%	>300M	Extremely high cost
MobileNet (Howard, 2017)	94–97%	<5M	Not medically optimised

2.4 Conclusion on Novelty

The title's use of 'Novel' is objectively justified. The thesis presents three original architectural modules (MAFE, MMSP, CAIP) that address three distinct gaps in the published literature, demonstrated through gap analysis tables, ablation experiments, cross-dataset validation, and comparative results against ten competing algorithms. The combination of highest accuracy (99.34%) with lowest parameter count (659K) across all reviewed methods constitutes a genuine and verifiable novel contribution to the field.

3. Committee Discussion Questions & Model Answers

The following questions cover all standard dimensions of a Master's thesis defence: motivation, methodology, architecture, results, limitations, and future work. Answers are written to be clear, confident, and academically precise.

Category A — Motivation & Problem Statement

Q1: Why did you choose brain tumor classification specifically? Why not another medical imaging task?
Answer: Brain tumors represent a critical diagnostic challenge because the four principal classes — glioma, meningioma, pituitary adenoma, and normal tissue — are morphologically diverse yet visually similar in MRI scans. Misclassification carries severe clinical consequences: wrong treatment selection, delayed therapy, or unnecessary chemotherapy. Moreover, there is a global shortage of neuroradiological expertise, particularly in low- and middle-income countries. This combination of high clinical stakes and under-resourced healthcare systems made brain tumor MRI classification the ideal testbed for an ultra-lightweight, deployable

model. The Mendeley dataset with 12,064 labelled images across four classes also offered a large, standardized benchmark enabling rigorous evaluation.

Q2: Why is a lightweight model necessary? Can we not just use a more powerful server for inference?

Answer:

Server-based inference assumes reliable internet connectivity, which does not exist in rural hospitals across Iraq and many developing nations. The clinical motivation is edge deployment: a model that runs on a tablet or basic diagnostic workstation without a cloud connection. The proposed model achieves 12 ms inference latency, 2.6 MB model size, and 4.4 MB RAM usage — all consistent with mobile and embedded device constraints. This was a deliberate design target, not an afterthought. Furthermore, reducing model size also reduces the risk of overfitting and lowers the carbon footprint of deployment, which are secondary but meaningful benefits.

Q3: What specific problem does your thesis solve that existing models do not?

Answer:

No prior published architecture simultaneously achieves above 99% classification accuracy and a model size below 3 MB on the brain tumor MRI classification task. The closest competitor, Khan and Auvée (2024), achieved 98.67% with 1.1M parameters — below both our accuracy threshold (99%) and our parameter budget (659K). This accuracy-efficiency gap is the central problem. Additionally, existing attention mechanisms (SE Networks, CBAM) were designed for general-purpose vision and are either channel-only or sequential in their attention fusion — neither is co-designed within a strict micro-parameter budget for medical imaging. Our MAFE module fills this architecture-level gap. Finally, no lightweight architecture prior to this work integrated a content-adaptive sigmoid-gated pooling layer within sub-700K parameters, which CAIP addresses.

Category B — Methodology & Architecture

Q4: Can you explain in simple terms how MAFE, MMSP, and CAIP each work and why they are arranged in this order?

Answer:

MAFE (Micro Adaptive Feature Extractor) comes first because it refines raw convolutional features by simultaneously identifying where tumors are located (spatial attention) and which feature channels carry the most diagnostic signal (channel attention). This is analogous to a radiologist first learning to focus on the right region and the right tissue characteristics before

doing detailed analysis. MMSP (Micro Multi-Scale Processor) follows because, once relevant features are selected, it processes them at three scales — fine detail, edge/boundary, and global context — to capture the full morphological picture of a tumor. This mimics a radiologist zooming in and out. CAIP (Context-Aware Intelligent Pooling) comes last to intelligently aggregate the multi-scale features into a compact representation for classification. Standard pooling either over-smooths (average) or over-sharpens (max); CAIP uses a sigmoid gate to adaptively blend both depending on the image content, preserving critical boundary information while suppressing noise.

Q5: What is the difference between your MAFE module and existing attention mechanisms like SE Networks or CBAM?

Answer:

SE Networks (Hu et al., 2018) perform channel recalibration only — they tell the model which feature channels to emphasize but say nothing about spatial location. CBAM (Woo et al., 2018) adds spatial attention but executes channel and spatial attention sequentially, meaning the spatial information is computed after channel weighting rather than simultaneously. MAFE performs both in parallel within a single module, requiring only approximately 8,000 parameters. This parallel co-design is architecturally unique and is specifically optimized for the micro-budget constraint of sub-700K total parameters. The clinical significance is that MAFE can simultaneously focus on the anatomical location of the tumor (spatial) and the most relevant MRI signal characteristics (channel) in one efficient step, which the sequential designs of CBAM and the single-mode design of SE cannot achieve.

Q6: How did you validate that each module genuinely contributes to the performance? Could the results simply come from one of the three modules?

Answer:

This concern is precisely what the ablation study in Section 4.11 addresses. I conducted a systematic experiment removing each module independently and in combination. Results show: the baseline model without any of the three modules achieves lower accuracy. Adding MAFE alone improves accuracy by a measurable margin. Adding MMSP alone provides a different improvement profile. Adding CAIP alone contributes +2.18% accuracy improvement — the strongest individual contribution. When all three modules are combined, the accuracy reaches 99.34%, which is greater than the sum of individual improvements, indicating positive synergistic interaction. This ablation evidence confirms that each module has a unique, non-redundant contribution to the final performance.

Q7: You used the Mendeley Brain Tumor MRI dataset. Why this dataset? Are there limitations in its use?

Answer:

The Mendeley dataset contains 12,064 high-resolution MRI images across four clinically relevant classes, making it the largest and most widely used benchmark for this task. Its use allows direct comparison with at least eight prior published methods (Table 2.1), which is essential for objective evaluation. However, I transparently disclose two limitations: First, the dataset uses image-level splitting (train/validation splits are random across all images from all patients), not patient-level splitting. In a real clinical setting, all images from one patient should be in the same split to avoid data leakage. Second, the dataset was collected from a single source with specific MRI scanner characteristics, which may not fully represent global MRI hardware diversity. The cross-dataset generalization experiments on Kaggle (99.24%), Br35H (98.33%), and Figshare (87.47%) partially address this by testing on independently collected data.

Q8: Why did you choose the Adam optimizer and what learning rate scheduling did you use?

Answer:

Adam was selected because it adapts the learning rate per parameter using first and second moment estimates, making it robust to sparse gradients and appropriate for the relatively small dataset size (12,064 images). A fixed learning rate was used with careful selection based on validation loss monitoring. The training configuration also included dropout regularization and batch normalization within the architecture to prevent overfitting. The training reproducibility was validated across three independent runs (Section 4.9), showing mean accuracy of 99.34% with low standard deviation, confirming that the results are not an artefact of a lucky initialization.

Category C — Results & Evaluation

Q9: Your model achieves 99.34% accuracy. Is this realistic, or might there be overfitting?

Answer:

Three mechanisms guard against this concern. First, the model uses dropout and batch normalization throughout its architecture, which are standard regularization techniques. Second, the performance was validated on three completely independent external datasets — Kaggle (99.24%), Br35H (98.33%), and Figshare (87.47%) — none of which were used during training or hyperparameter tuning. The drop to 87.47% on Figshare is expected and acknowledged: Figshare uses a different scanner with lower resolution and different

annotation protocols, demonstrating that the model is sensitive to distribution shift rather than inflated. Third, three independent training runs from different random initializations were conducted, showing consistent results (Section 4.9), which rules out lucky initialization. The 99.34% figure is therefore credible and defensible.

Q10: The Figshare cross-dataset accuracy drops to 87.47%. Is this not a significant weakness?

Answer:

The Figshare drop is not a weakness to be hidden — it is an important, transparent finding that I report explicitly. Figshare uses lower-resolution MRI images from a different scanner and has different class definitions compared to the training data. The 11–12% accuracy gap reflects genuine domain shift, which is a known challenge in medical AI. The key point is that the model still achieves 87.47% on a domain-shifted dataset while having been trained only on Mendeley data — demonstrating meaningful generalization. For comparison, many prior state-of-the-art methods in the literature do not report any cross-dataset results at all, meaning their generalization is completely unknown. Our transparent reporting of this limitation, including the likely cause (scanner and resolution differences), is scientifically rigorous.

Q11: You compare against 10 competing algorithms. Is this comparison fair given different training conditions?

Answer:

This is a valid concern that I address explicitly in the thesis. The comparative analysis in the 20-epoch rapid-training protocol (Section 4.18) ensures all ten algorithms are trained under identical conditions: same dataset split, same hardware environment (NVIDIA GPU with CUDA), same number of epochs, and same evaluation protocol. This controls for training duration and data differences. For the published literature comparison (Table 4.7), results are taken directly from the original papers and compared on the same dataset, following standard academic practice. I acknowledge that published results may use different hyperparameter tuning budgets, which is an inherent limitation of cross-paper comparisons that I disclose in the discussion section.

Q12: What do the Grad-CAM visualizations (Section 4.15) tell us, and why are they important?

Answer:

Grad-CAM (Gradient-weighted Class Activation Mapping) produces heatmaps showing which regions of the input MRI image most strongly influence the model's classification decision. In

Section 4.15, the Grad-CAM visualizations demonstrate that the model focuses on anatomically correct regions — the tumor mass itself — rather than background noise, scanner artifacts, or surrounding healthy tissue. This is critical for clinical trust: a model that achieves 99.34% accuracy by exploiting spurious correlations (e.g., scanner watermarks or patient metadata) would be clinically dangerous. The Grad-CAM evidence supports that the model has learned genuine tumor morphology, providing an interpretability foundation for clinical deployment. This is also why I selected Grad-CAM over simple accuracy — explainability is a prerequisite for medical AI adoption.

Category D — Limitations & Critical Questions

Q13: You mention patient-level vs. image-level splitting as a limitation. How serious is this and why did you not use patient-level splitting?

Answer:

Patient-level splitting is the gold standard for clinical AI validation because it ensures that images from the same patient never appear in both training and validation sets, preventing information leakage. Image-level splitting, as used in this thesis, may allow slight leakage if one patient's images appear in both sets, potentially inflating the reported accuracy. I disclose this limitation explicitly in Section 4.5. The reason for not using patient-level splitting is that the Mendeley dataset does not provide patient identifiers — each image is independently labelled without linking images to specific patients. This is a limitation of the dataset itself, not a methodological choice. I recommend that future work use datasets with explicit patient metadata to enable proper patient-level splitting.

Q14: Your model has not been tested in a real clinical environment. Can we trust the results?

Answer:

The results are laboratory results, and I state this clearly. The model has been tested on benchmark datasets following standard machine learning evaluation protocols, which is the appropriate first step before clinical deployment. Moving from laboratory to clinical deployment requires additional steps: regulatory approval, prospective clinical trials with radiologist comparison studies, integration testing with hospital PACS (Picture Archiving and Communication Systems), and patient data privacy compliance. These are beyond the scope of a Master's thesis but are identified as future work recommendations. What the thesis does establish is a strong proof-of-concept with rigorous multi-dataset validation, which is the necessary prerequisite for subsequent clinical translation.

Q15: Could a transformer-based model (like Vision Transformer) outperform your CNN-based approach?

Answer:

Vision Transformers (ViT) achieve state-of-the-art performance on general vision tasks but carry substantial computational costs — the ensemble ViT used by Tummala et al. (2022) has over 300 million parameters. For edge deployment on mobile medical devices with 2.6 MB model size and 4.4 MB RAM constraints, a full ViT is completely infeasible. The MMSP module in this thesis incorporates a lightweight self-attention-inspired mechanism through its multi-scale fusion, capturing some benefits of multi-head attention without the quadratic complexity cost. Future work could explore hybrid lightweight transformer-CNN architectures, which is an active and promising research direction. However, within the strict computational budget defined by this thesis, the Modified CNN architecture represents the current optimal solution.

Q16: Why did you not use transfer learning from ImageNet pre-trained models?

Answer:

Transfer learning from ImageNet pre-trained models (e.g., VGG16, ResNet, EfficientNet) introduces two problems for this specific application. First, ImageNet models are trained on natural photographs (dogs, cars, furniture) whose low-level feature statistics differ substantially from MRI grayscale medical images, raising questions about transfer validity. Second, large pre-trained models inherently carry millions of fixed or fine-tuned parameters that prevent achieving the sub-700K parameter budget required for mobile deployment — EfficientNet-B0 alone has 5.3M parameters. The proposed framework is designed from scratch and trained from random initialization on domain-specific MRI data, which avoids domain mismatch and respects the deployment constraint. The ablation study confirms that the three novel modules are sufficient to reach 99.34% without any pre-training.

Category E — Future Work & Big Picture

Q17: What are the most important future research directions you would pursue after this thesis?

Answer:

Four directions are prioritized: First, patient-level dataset collection with proper patient identifiers to enable rigorous patient-level train-validation splits and eliminate the image-level leakage limitation. Second, prospective clinical validation in collaboration with Wasit Teaching Hospital or equivalent institution, comparing model decisions against radiologist interpretations on real patients. Third, extension to MRI segmentation (not just classification) — identifying the precise spatial boundary of the tumor in addition to its type — using the MAFE and MMSP modules as a lightweight encoder within a U-Net-style decoder. Fourth,

federated learning implementation to train the model across multiple hospitals without sharing patient data, addressing privacy compliance while improving generalization across scanner types.

Q18: If you were to redo this thesis, what would you change?

Answer:

Three things. First, I would prioritize obtaining a dataset with patient identifiers from the beginning to perform proper patient-level splitting. Second, I would include a clinician evaluation component — having a radiologist review Grad-CAM outputs and classify a sample of images independently, then comparing model accuracy vs. human accuracy. This would provide a clinical benchmark rather than purely algorithmic comparison. Third, I would explore semi-supervised learning techniques to make better use of the many unlabelled MRI images available in clinical systems — labelled medical data is scarce and expensive, and the current model uses only labelled data.

Q19: What is the real-world impact of this thesis if the model were deployed?

Answer:

If deployed in a low-resource clinical setting — a district hospital in Iraq, for example — a radiologist or general practitioner could run the model on a basic tablet or laptop without internet connectivity. A patient presenting with neurological symptoms would receive an MRI scan; the model provides a preliminary classification in 12 ms, flagging suspected tumor class for radiologist review. This does not replace the radiologist — it acts as an AI-assisted triage and second-opinion tool, reducing diagnostic delays and workload. In settings where a neuroradiologist is unavailable (which describes most district hospitals in Iraq and across the developing world), this could directly reduce the time between imaging and treatment initiation, which is a critical prognostic factor in high-grade glioma where survival is measured in months.

Q20: Is the word 'Novel' in your title justified? What exactly is new about your work?

Answer:

Yes — and I can answer this specifically. Three things are genuinely new. First, the MAFE module: no prior published work has implemented a parallel (not sequential) dual spatial-channel attention mechanism co-designed within a sub-700K parameter budget for brain tumor MRI classification. Second, the CAIP module: no prior lightweight architecture integrates a sigmoid-gated content-adaptive pooling within this parameter constraint. Third, the simultaneous achievement: no prior work achieves both above 99% accuracy and below 3 MB

model size on this task. Each of these claims is supported by citation-backed gap analysis tables (Table 2.2), direct performance comparisons (Table 4.7), and ablation experiments (Table 4.6) that demonstrate the unique contribution of each module. The novelty is architectural, empirical, and clinically motivated — all three dimensions that a Master's thesis novelty claim requires.

4. Key Numbers — Quick Reference for Defence

Memorize these figures. The committee will test them.

Metric	Value	Significance
Primary dataset	Mendeley — 12,064 MRI images	4 classes: glioma, meningioma, pituitary, no-tumor
Total parameters	659,228 (659K)	Lowest among all compared methods
Model file size	2.6 MB	< 3 MB mobile deployment target — achieved
RAM usage at inference	4.4 MB	Compatible with entry-level tablets
Inference latency	12 ms	Real-time clinical use
Primary accuracy	99.34%	Highest among all compared methods
Kaggle (external)	99.24%	Excellent cross-dataset generalization
Br35H (external)	98.33%	Good cross-dataset generalization
Figshare (external)	87.47%	Domain shift — lower res scanner, acknowledged
CAIP ablation contribution	+2.18% accuracy	Strongest single-module contribution
Parameter savings vs U-Net	80% fewer params	vs. >3.2M parameters in comparable U-Net methods
Training runs for stats	3 independent runs	Confirms reproducibility, not lucky init
Competing algorithms tested	10 total	Deep CNN, VGG16, ResNet18, EfficientNet, MobileNetV2, LSTM, XGBoost, RF, SVM

Key Terms to Define Confidently

Term	One-line Definition
MAFE	Parallel spatial + channel attention module — ~8K params — unique co-design within micro-budget
MMSP	Three-scale feature processor (fine / boundary / global) using DWConv + learnable fusion

Term	One-line Definition
CAIP	Sigmoid-gated adaptive blending of max and average pooling — content-aware, not static
Ablation study	Removing each module independently to verify its unique contribution to performance
Grad-CAM	Heatmap showing which image regions drive classification — interpretability proof
Patient-level split	Train/validation division ensuring no patient's images appear in both sets
Depthwise conv	Convolution applied channel-by-channel — reduces parameters without losing feature coverage
McNemar's test	Statistical significance test comparing two classifiers on matched pairs of predictions
